10Alytics

Specialization: *Exploratory Data Analysis*

Business Focus: *Aviation*

Tool: *Pandas, Matplotlib, Seaborn.*

Case Study: Aviation Sector Insights

Project Learning Opportunities

- ***Analyze Performance:*** *Assess airline operations, including delays and passenger satisfaction.*
- ***Explore Correlations:*** *Uncover relationships between variables such as weather, delays, and revenue.*
- ***Visualize Trends:*** *Apply advanced visualization techniques for storytelling.*

Tools and Technology to be Used









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Background

- SkyBound Analytics is a data consulting firm partnering with major airlines globally to improve operational efficiency and customer satisfaction.
- With the aviation industry rapidly evolving, airlines are competing on minimizing delays, enhancing passenger experience, and optimizing profitability.
- To tackle challenges such as flight delays, fluctuating customer satisfaction, and varying profitability across routes, SkyBound Analytics has obtained operational data from five major airlines for a detailed exploratory data analysis.



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Problem Statement



Airlines are struggling with multiple operational challenges:

- **Operational Inefficiencies:** Frequent delays attributed to weather, aircraft types, and peak travel seasons.
- **Customer Experience:** Declining satisfaction scores linked to long delays and inconsistent loyalty program benefits.
- **Revenue Optimization:** Understanding how ticket class, seasonality, and customer demographics influence revenue generation.
- **Performance Analysis:** Identifying which airlines, aircraft types, or regions underperform in terms of delay times and passenger numbers.

To address these challenges, SkyBound Analytics will leverage EDA to uncover patterns and provide actionable recommendations for airline partners.



Rationale

- In a competitive aviation industry, airlines face constant pressure to optimize their operations and meet customer expectations.
- Delays, poor service quality, and inefficient route planning can negatively impact reputation and revenue. This case study aims to uncover insights through EDA to:
  - ***Improve Operational Efficiency:*** By analyzing delay causes and mitigating risks associated with weather and route scheduling.
  - ***Maximize Revenue:*** Through insights into seasonal trends, flight classes, and passenger behavior.
  - ***Enhance Customer Experience:*** By identifying and addressing pain points that impact satisfaction.



Objectives

The primary objectives of this case study are to:

1. ***Understand Key Factors Influencing Airline Operations:*** Identify and analyze factors such as delays, weather conditions, and customer satisfaction that impact airline performance.
2. ***Evaluate Revenue Drivers:*** Determine how passenger count, flight duration, flight class, and ticket prices contribute to overall revenue.
3. ***Analyze Operational Efficiency:*** Examine correlations between delays, fuel costs, and other operational metrics to identify areas for improvement.
4. ***Explore Customer Satisfaction:*** Investigate how various factors like delays, airline service quality, and flight class impact customer satisfaction scores.
5. ***Develop Actionable Recommendations:*** Use insights from the analysis to provide recommendations for improving operations, increasing revenue, and enhancing customer experience.

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Data Dictionary

Column Name	Description	Type
Date	Flight date	DateTime
Airline	Name of the airline (e.g., BlueJet, SkyTrack, Sunshine, etc.)	Categorical
AircraftType	Type of aircraft used (e.g., Boeing 787, Airbus A380, etc)	Categorical
FlightDuration	Duration of flight in minutes	Numerical
Passengers	Number of passengers per flight	Numerical
FlightClass	Class of ticket (Economy, Business, First)	Categorical
DestinationRegion	Region of destination (e.g., Domestic, International)	Categorical
WeatherCondition	Weather conditions during the flight (e.g., Clear, Rainy)	Categorical
FuelCost	Fuel cost per flight in USD	Numerical
DelayMinutes	Delay duration in minutes	Numerical
CustomerSatisfaction	Customer satisfaction score	Numerical
LoyaltyProgram	Whether the passenger is enrolled in a loyalty program	Binary
Weekday	Day of the week the flight operated	Categorical
PeakSeason	Whether the flight occurred during a peak travel season	Binary
TicketType	Type of ticket purchased (Refundable, Non-Refundable)	Categorical

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Project Workflow

STEP 1

Data Cleaning:
- *Identify and handle missing values.*
- *Remove duplicate records.*
- *Address outliers in numerical columns*

STEP 2

Basic Data Exploration:
- *Check the structure and summary of the dataset* ***(.info(), .describe())***.

STEP 3

Univariate Analysis:
- *Visualize distributions of numerical data to confirm outliers.*
- *Count plots for categorical data.*

STEP 4

Bivariate Analysis:
- *Identify relationships between two (numerical vs numerical, numerical vs categorical, and categorical vs categorical) variables.*

STEP 5

Multivariate Analysis:
- *Identify relationships between three or more variables.*

STEP 6

Feature Engineering:
- *Create new columns like:*
  - ***Revenue per Passenger****: Revenue / Passengers.*
  - ***Delay Rate****: Ratio of DelayMinutes to FlightDuration.*
  - *Categorize DelayMinutes into bins (e.g., No Delay, Minor Delay, Major Delay).*
- *Encode categorical variables for modeling (if required later).*

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READY TO DELVE IN?

